

Personal Profile

Final-year Electrical Engineering student at Sharif University of Technology with a software focus, applying for a research-oriented MSc in computer science, software systems or data-driven systems, and open to assistant roles in research or teaching. Strong results in programming, embedded systems and data science (all 19/20 or above), four consecutive semesters of teaching assistance in programming courses, a bot platform adopted by a second university, and publicly documented project code.

Education

2023–2027 **Sharif University of Technology, Tehran, Iran**

B.Sc., Electrical Engineering (Digital Systems focus) • expected Aug 2027

- **Cumulative GPA: 16.88/20.** Above the department average (16.49) and university average (16.09) for this cohort, as printed on the [Sharif transcript](#).
- Most recent semester (Spring 2026): **19.30/20** over 19 units.
- **National rank 134** among about 150,000 candidates in the Mathematics and Physics group of the Iranian University Entrance Exam (Konkur).
- **Relevant coursework (grade/20):** Fundamentals of Programming (20.0); Foundations of Data Science (19.3); Embedded Systems (19.8); Object-Oriented Programming (17.8); Computer Interfacing, a graduate course (19.0); Computer Structure and Microprocessor with lab (17.0). In progress (Fall 2026): Parallel Programming; Data Structures and Algorithms.

Projects

2026–present **SUT Music: Data Platform and Recommendation Engine**

[source](#)

Foundations of Data Science course, Dr. Salavati • Python, scikit-learn, PostgreSQL, FastAPI, React

- **Data:** Led the scraping and cleaning of a Telegram music community's history into a database of 14,800+ tracks and 86,900+ reactions.
- **Recommenders:** Built two generations (matrix factorization, then a two-stage artist-to-track ensemble) behind a backend API and a Telegram Web App.
- **Team:** Team of 3; project lead.

2024–present **EE Department Bot Ecosystem and BotOS Framework**

Lead developer • Python (asyncio), PostgreSQL, Docker, Google Sheets API

- **System:** Built and maintain bots for course-file distribution, assignment grading and re-judge ticketing, used by the EE department and by two student groups.
- **Framework:** Designed BotOS, a framework that deploys new bots from declarative JSON configuration in isolated Docker containers.
- **Adoption:** Began in the programming courses I assisted (Fundamentals of Programming and OOP); the platform has since been adopted by a second university (IUST).

Spring 2026 **Smart Surveillance System on an Embedded Linux Board**

[source](#)

Embedded Systems course, Dr. Gholampour • Orange Pi Zero Plus 2 (H5), C, Python

- **System:** Individual project built around an on-board person-detection system in which a Haar-cascade detector feeds an HTTPS dashboard, a REST API and an MQTT/e-mail alert daemon.
- **Reliability:** Added a software watchdog that restarts the detector when the camera feed stalls, hysteresis-based thermal throttling and an event log.
- **Evaluation:** Measured boot time, detection accuracy under varied lighting, concurrent load and temperature with and without a fan; documented in a technical report.

Spring 2026 **OTP Memory Controller (RTL design, synthesis and verification)**

[source](#)

ASIC/FPGA System Design course, Dr. Baharvand • Verilog, Cadence Genus and Xcelium

- **Design:** Co-designed a power-on FSM controller that reads a 256-bit OTP array, applies hardware patches and checks a CRC-16.
- **Team and flow:** Team of 4, led week 1 (rotating lead); verified with a testbench, then ran synthesis and gate-level simulation.

Spring 2025 **MIPS32 Processor Implementation on FPGA**

Computer Architecture lab, Dr. Movahedin • Verilog, MIPS32 assembly, Xilinx Zynq-7010

- **Design and validation:** Single-cycle, multi-cycle and pipelined MIPS32 CPUs; the multi-cycle one ran on an FPGA board.

Teaching Experience

- 2024–2026 **Teaching Assistant, Fundamentals of Programming and OOP**
Sharif University of Technology • Dr. Vosoughi Vahdat and Dr. Arasteh • four consecutive semesters
- Fall 2025 **Teaching Assistant, Electronics 1**
Sharif University of Technology • Dr. Fakharzadeh, Dr. Sarvari, Dr. Medi, Dr. Besharatirad
- Fall 2025 **Lab Assistant, Logic Circuit Lab (FPGA)**
Sharif University of Technology • Dr. Movahedin • supported students designing on Zynq-7010 FPGA boards
- Spring 2026 **Teaching Assistant, Theory of Circuits**
Iran University of Science and Technology (IUST)

Industrial Training

- Summer 2026 **Trainee, Parsian Zarin (ABLY): MQL5 expert advisors; Excellent Pass**

Technical Skills

- **Languages:** Python, C, C++, Java, SQL, Verilog, MIPS32 assembly.
- **Software engineering:** backend and REST API development, asynchronous programming, Docker, Git, CI/CD, Linux, relational databases (PostgreSQL, SQLite), React (web front end).
- **Data and machine learning:** data analysis, machine learning, recommender systems.
- **Embedded and hardware:** embedded Linux on single-board computers, FPGA design with Xilinx Vivado, Cadence Genus and Xcelium.

Languages

Persian (native); English (official test score pending; CEFR level to be confirmed by the test report).